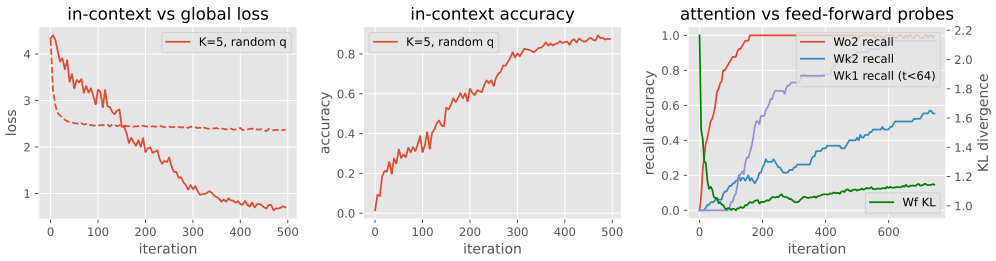

Figure 10: Training of a more realistic architecture with (i) ReLU MLP instead of linear layer for the second feed-forward layer, (ii) all parameters trained, including embeddings, (iii) pre-layer normalization. The loss, in-context accuracy and memory recall probes are similar to the simplified architecture (see, *e.g.*, Figure 4).

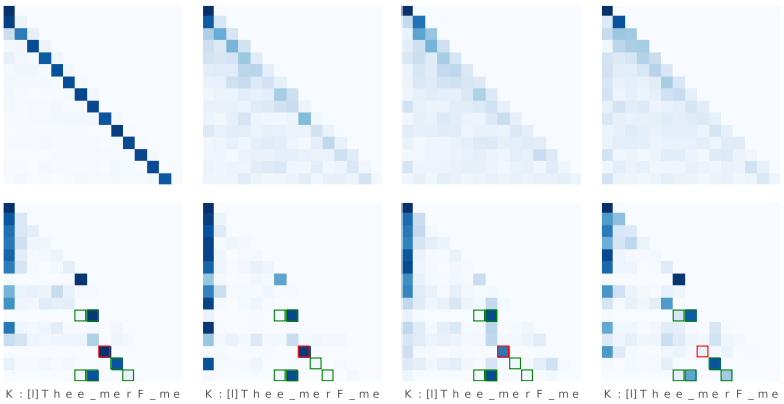

Figure 11: Attention maps for a two-layer model with 4 attention heads. In the first layer (top), the previous token mechanism is mostly achieved by one of the four heads, while the induction behavior at the second layer (bottom) is distributed across the different heads.

attention input to W_O^2, given by

$$R_1 = \frac{1}{N}\sum_{k=1}^{N} \mathbb{1}\left\{k = \arg\max_{k'} (W_V^2 w_E(k'))^\top (\mu_k - \mu)\right\},$$

where $\mu_k = \mathbb{E}[x|y=k]$ and $\mu = \mathbb{E}[x]$, for $x = \frac{1}{t}\sum_{s=1}^{t} W_V^2 w_E(z_s)$ and $y = z_{t+1}$, when z_t is a trigger token after its first occurrence. Expectations are computed over batches of data of varying sizes and in different dimensions. We call this "one-step" since it is related to the classifier obtained after performing a single gradient step on W_O^2 from zero initialization (see Lemma 2 and Appendix B.3.1). The plots illustrate that this simple one-step model is already able to extract relevant signal from the noisy average attention, after a handful of batches of data, corresponding to tens of thousands of tokens, and that this gets easier as the dimension increases.

More complex architectures. Figure 10 shows training behavior for a more complex model than the simplified one considered in Section 5, namely where we train all parameters, replace the linear W_F feedforward layer by a two-layer MLP, and were (pre-)layer-normalization is added. Despite these changes, we see similar behavior for the memory recall probes (which now involve embeddings that may change over time), suggesting that the model is still identifying the same memory associations, despite the additional redundancies in parameters and modified training dynamics.

Figure 11 shows the attention maps obtained when training a multi-head version of our two-layer model, with four attention heads per layer. We see that the redundancy of multiple heads creates difficulties in identifiability: only one of the first layer heads learns the previous token behavior, while the induction behavior is shared across different heads at the second layer. This illustrates the challenges of interpretability in the presence of redundant models, which then require additional work to identify which of the layers and heads are performing a given behavior, *e.g.*, through interventions and causal mediation analysis [36, 56].